The Particle Model — Paper A

MEDIUM · NON-CALCULATOR · 35 MINUTES · 30 MARKS

NAME DATE SCORE / 30

- Answer every question in the space provided.
- No calculator is needed anywhere on this paper.
- Where a question says explain, your answer must talk about particles — their arrangement, their movement, the spaces between them, or the forces between them.
- The mark for each question is shown in square brackets on the right.

SECTION 1 — THE THREE STATES

1. Complete the table. Each empty box is worth one mark. [6]

	Solid	**Liquid**	**Gas**
Arrangement	regular pattern		
Movement		slide past each other	
Spacing	very close		

2. Name the change of state in each case.

(a) Solid to liquid [1]

(b) Gas to liquid [1]

(c) Liquid to gas [1]

(d) Solid straight to gas [1]

3. Answer these two questions about the properties of the states.

(a) Which state has a fixed volume but no fixed shape? [1]

(b) Which state has neither a fixed volume nor a fixed shape? [1]

SECTION 2 — DIAGRAMS AND EXPLANATIONS

4. In the space below, draw a particle diagram for a solid, a liquid and a gas. Label each one. [3]

5. Explain why a gas can easily be compressed, but a liquid can hardly be compressed at all. [3]

6. State the melting point and the boiling point of water in degrees Celsius. [2]

7. Explain, in terms of particles, why a solid keeps its shape. [2]

8. Changes of state are described as physical changes rather than chemical changes. State what this means, and give one piece of evidence for it. [3]

9. 20 g of ice is left in a sealed container until it has all melted. State the mass of water produced, and explain your answer. [2]

10. Give three properties of a gas. [3]

End of paper. Check every answer has a unit or a form the question actually asked for.